



Wibral PID research status

Mathematical progress and unresolved verification findings

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This private report records mathematical progress and the remaining checks. It does not announce a new formally verified theorem or application benefit. Remote main was read back at commit `f93f2b6` on 30 September. Its recorded CI observation at 22:14 UTC found all 47 jobs successful. These checks cover that committed software, not the unfinished proofs or later verifier drafts.

Mathematical scope and progress

The object is the categorical shared-exclusions PID of Makkeh, Gutknecht and Wibral, not another PID functional or the continuous estimator. Assume one normalized law on finite source and target alphabets, at least two sources, positive-probability anchors, the full antichain lattice and natural-log units (nats). The argument below concerns exact probabilities. It is not an estimator calibration claim; independent sampling is not needed for the finite-law identity.

Each lattice node is a nonempty family of nonempty source sets, with no member containing another. The order $\beta \preceq \alpha$ means that every set in α contains some set in β . At an anchor, a node's event is the OR of its sets' matching-source AND events. Its informative component is minus the logarithm of that event's probability; its misinformative component uses the probability conditional on the target. The component atoms are the unique coefficients whose sums below each node recover these components. Write π_β^+, π_β^- for their averages under the common law.

For a nonempty proper retained source group C , write R for the other sources. Let K be exactly the set of nodes not below or equal to the single-set node $\{C\}$. Define $A = H(S_R | S_C)$, $B = H(S_R | S_C, T)$ and $\Delta = I(S_R; T | S_C) = A - B$. Subtracting the component value at $\{C\}$ from its top-node value removes exactly the atoms below $\{C\}$. Averaging the remaining joint-minus-retained surprisal gives conditional entropy. Thus the manually reviewed derivation gives component budgets $\sum_{\beta \in K} \pi_\beta^+ = A$ and $\sum_{\beta \in K} \pi_\beta^- = B$. Using the published nonnegativity of these exact component atoms, the absolute **averaged** signed mass $M = \sum_{\beta \in K} |\pi_\beta^+ - \pi_\beta^-|$ obeys

$$\Delta \leq M \leq A + B = 2A - \Delta.$$

The triangle inequality gives the lower bound; component nonnegativity gives the upper bound. This is not the average of absolute pointwise atoms. A separate event-based proof route seeks to derive positivity from actual probabilities; its convergence, lattice and formal-checking obligations remain open.

The question matters when a sensor or neural source may be redundant. If $S_R = f(S_C)$ on the law's positive support, the manual event argument predicts zero components at every position in K . Observed duplicate rows alone cannot establish this population premise. MI, conditional MI and direct task loss are the first baselines when only prediction or source removal is at issue.





Negative example. Let $S_1 = X$, $S_2 = Z$ and $T = X$, with independent fair bits X, Z . Here $C = \{1\}$, $R = \{2\}$ and K consists of the two nodes $\{\{2\}\}$ and $\{\{1, 2\}\}$. The four joint states each have probability $1/4$. For a fixed anchor, $P(S_1 = x \text{ or } S_2 = z) = 3/4$, whereas the same event has probability one given $T = x$. Thus redundancy is $r = \log(4/3)$. Since $I(S_2; T) = 0$ and $I(S_1; T) = I(S_1, S_2; T) = \log 2$, source 2 unique information is $-r$ and synergy is r . Hence $\Delta = 0$ but $M = 2 \log(4/3) > 0$. Zero conditional MI does not force individual PID atoms to vanish.

Five unresolved verifier findings

Independent source review found the following causes in the stopped v5 verifier. These are static findings, not reproduced runtime failures. They do not establish a positive-selection bypass or a counterexample to the published PID.

1. **Early persistence failure.** A directory synchronization error can skip feasible original evidence capture. Start failure records before persistence and attempt each safe role.
2. **Lost input prefixes.** A later read error can discard earlier measured pins and shrink later checking. Retain prefixes and derive the complete safe recheck domain separately.
3. **Incomplete negative guards.** A trusted negative can retain semantic facts that its producer could not reach. Check producer identity, stage and the exact semantic-fact census.
4. **Lost receipt prefixes.** A later error can discard earlier receipt pins, including an observed result. Keep per-file outcomes and prior pins; do not invent missing receipts.
5. **Suppressed role rechecks.** One unavailable original role can prevent checking other available roles. Recheck each safe role independently and retain actual drift evidence.

The repair design is independently reviewed. A new v6 draft has source-writing permission only; the worker reported at 22:57 UTC that no v6 files or tests had been produced. This is a dated author report, not a verified repair. Imports, native calls and formal acceptance are not authorized by that permission. The separate fixture-guard conflict remains unresolved. All 19 Fin3 mathematical targets remain open. No guard, limit, kernel flag or numerical tolerance is relaxed to obtain a pass.

Application and preservation limits

The proposed neural pilot remains exactly 32 reset trials: 16 training and 16 sealed confirmation trials. Its harness, seed and split manifest, installed runtime qualification and editor launch window are not ready. The exposed five-row simulator law is descriptive evidence, not held-out prediction or physical/biological benefit. No smaller substitute experiment was run.

All nine named iCloud roots remain protected. One complete local recovery does not supply approved distinct custody or deletion clearance; other roots retain missing-byte or revision gaps. Failed attempts, counterexamples and source preimages are kept. No other-editor files or cloud roots are removed.





Rust implementation and computational cost

At main revision `f93f2b6`, the default-build categorical API includes `stable::categorical::discrete_sxpid_n_averaged_with_budget_and_cancellation`, implemented in `crates/pid-core/src/sxpid.rs`. It accepts two to four source matrices and a target matrix. Write N for aligned rows, D for their total column width and Q for occupied complete source-target states. This report adds no Rust API, continuous estimator or deployed learner. The general-coalition budget wrapper and event-based formal route are not implemented or benchmarked. For already computed averaged atoms, summing the $|K|$ absolute values requires $O(|K|)$ arithmetic and $O(1)$ extra accumulator space. This is an arithmetic-operation count, not a timing or numerical-error guarantee; it excludes estimating the law and computing its atoms. Inputs to an eventual empirical wrapper would be categorical rows with explicit source and target alphabets and occupied support. Sampling, resampling, cancellation, resource limits and binary64 error bounds still require their own contracts. No timing, calibration or real-time suitability is claimed.

Evidence and next checks

The six-page *Source redundancy and SxPID3 atom budgets* note gives the detailed three-source derivation and negative examples. The companion evidence map names its exact local lane. The general-law design remains a manual proof proposal. Conditional linear SMT checks are not probability proofs. Next checks are the exact v6 source review and hostile controls; formal event, logarithm and lattice correspondence; numerical refinement; and the unchanged neural pilot admission. Same-host reviews and sequential hashes do not prove atomic custody or independent archival preservation.

Defining source: Makkeh, Gutknecht and Wibral, *Introducing a differentiable measure of pointwise shared information*, Physical Review E 103, 032149 (2021), [arXiv version 5](#). The budgets and proposed proof routes are repository derivations, with no scientific-priority claim.

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